Lab Template Stat 240

Lab 02

AUTHOR
Peter Khrissanov

Problem 1

Part a)

```
median_func <- function(vector){

  sorted <- sort(vector)
  len <- length(sorted)
  if(len %% 2 == 0){
    num1 <- sorted[(len/ 2)+1]
    num2 <- sorted[len / 2 ]
    return ((num1+num2) / 2)

  }
  {
      val <- sorted[(len/2)+1]
      return (val)
  }

}

median_func(c(1,2,3,4, 5))
```

```
[1] 3
```

Part b)

```
## write answer here
a <- (301611434 %% 27) + 1
library(readr)
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ purrr     1.2.1
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.1     ✔ tibble    3.3.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.2
```

```
── Conflicts ──────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
become errors
```

```r
data <- read_tsv("AS-N100.tsv")
```

```
Rows: 10000 Columns: 10
── Column specification ────────────────────────────────────────────────────────
Delimiter: "\t"
chr  (1): ticker
dbl  (7): open, high, low, close, volume, quantity, tradecount
date (1): date
time (1): timestamp

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
symbols <- sort(unique(data$ticker))[a]

worked <- filter(data, ticker == symbols)
count <- c(worked$open)

median_func(count)
```

```
[1] 106.99
```

STAT 240 Lab D101

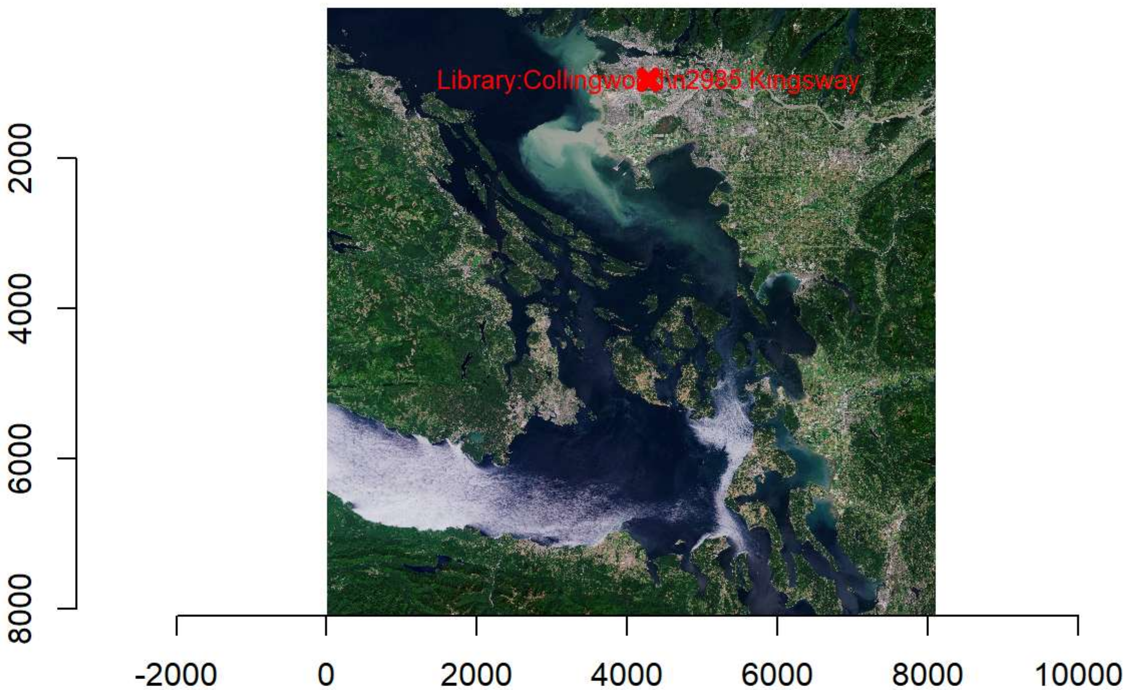
Lab 2

AUTHOR
Peter Khrissanov

Problem 2

Part 2)

```
library(rjson)
library(imager)
libraries <- fromJSON(file = "libraries.json")
image <- load.image("Figure03.png")


b <- (301611434 %% 21) + 1

cords <- libraries[["features"]][[b]][["geometry"]]$coordinates
lib <- libraries[["features"]][[b]][["properties"]]$maptip

lat_top <- 49.410705
lon_left <- -124.217671
lat_bottom <- 47.929083
lon_right <- -121.994887

lat <- cords[2]
lon <- cords[1]

W <- width(image)
H <- height(image)


x <- (lon - lon_left) / (lon_right - lon_left) * (W - 1 ) + 1
y <- (lat - lat_top) / (lat_bottom - lat_top) * (H - 1) + 1
plot(image, axes + FALSE)
points(x,y, col = "red", pch = 4, lwd = 5)
text(x, y + 18, lib, col="red", cex = 0.8)
```


Library:Collingwo
\n2985 Kingsway
2000
4000
6000
8000
-2000
0
2000
4000
6000
8000
10000